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DATA DESCRIPTOR

AgriVision: A Benchmark Dataset for Advancing Real-World Robotic Vision in Densely Fruited Blueberry Crop

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Robotic vision for densely fruited crop (i.e., blueberry) management remains challenging due to complex real-world conditions such as irregular fruit structures, overlapping clusters, varying berry sizes, inconsistent lighting, and cluttered backgrounds. These factors are compounded by the scarcity of diverse, high-quality annotated data from production environments, which is critical for training robust detection models. To address this gap, we introduce a comprehensive large-scale dataset repository for dense blueberry analysis, comprising three subsets aimed at advancing learning paradigms in this domain: DB-1) 1,195 fully annotated high-resolution images for supervised learning, DB-2) 141K frames from 520 videos for weakly/semi-supervised learning, and DB-3) 10K synthetic images with annotations, generated via our proposed data realization algorithm that mimics real-field complexity by offering a scalable, cost-effective foundation for blueberry annotation and robust model generalization. We validated the utility of DB-1 by benchmarking a strong baseline and proposing a customized framework for densely clustered blueberries, achieving 75.06% SEN, 56.85% IoU, and 72.49% DICE, outperforming the strongest baseline by 24.71%, 13.5%, and 8.6%, respectively. Further implementation and supplementary details are available at our GitHub repository: <https://github.com/Owais-CodeHub/PVT-SN-SAM-RN>.

Background & Summary

Robotic vision based on advanced artificial intelligence (AI) facilitates efficient estimation of crop yields, ripeness analysis, and disease detection, improves decision-making, and promotes sustainable agriculture^{1,2}. However, accurately detecting densely clustered crops such as blueberries for robotic vision applications is challenging due to factors such as occlusions, uneven fruit shapes, and overlapping clusters. These complexities are compounded in real-world agricultural environments, where variations in lighting, crop density, and complex backgrounds further hinder accurate segmentation. For accurate yield estimation and ripeness analysis, AI models must effectively distinguish individual fruits within dense clusters and adapt to various environmental conditions^{3,4}.

Recent advances in AI have significantly impacted precision agriculture, enabling novel solutions to complex challenges in crop segmentation. Deep learning models originally developed for general computer vision applications have been adapted for agricultural settings^{5–7}. For example, Abdalla *et al.*⁸ fine-tuned a convolutional neural network (CNN) for semantic segmentation of oil seed ravishment under high weed pressure. Although their method demonstrated promising results, its reliance on transfer learning limits its adaptability to new datasets with distinct characteristics. Peng *et al.*⁹ applied DeepLabv3+ (DLv3+) to segment the branches of litchi, focusing on tree structures. Although effective for this scenario, the model struggles with the occlusions and overlapping clusters characteristic of densely clustered blueberries. Similarly, Sun *et al.*¹⁰ used DLv3+¹¹ with shape constraint level sets for flower detection in apple, peach, and pear crops. This work highlighted the

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importance of geometric constraints, but was limited by the diversity of the dataset and the focus on flowering stages, reducing its applicability to dense blueberry segmentation.

Transformer-based architectures have also shown promise in precision agriculture. Zhang *et al.*¹² utilized SegFormer⁵ to extract cotton planting areas from high-resolution remote sensing images, demonstrating the scalability of transformer-based models. However, segmenting dense blueberry clusters remains challenging. Similarly, Xu *et al.*¹³ introduced the Multiscale Contextual Swin Transformer (MSC-Swin) for crop image segmentation, which effectively captured multiscale features. Although these transformer-based methods achieved strong performance in controlled environments, their generalizability to highly cluttered and occluded blueberry environments remains unexplored.

To address the need for specialized datasets, the FruitSeg30 dataset¹⁴ was developed, providing annotated images to benchmark fruit segmentation models. Using this dataset, a U-Net-based model was implemented, achieving satisfactory segmentation results. However, the dataset lacks the scale and complexity required for real-world agricultural challenges, such as the analysis of densely clustered blueberries. Efforts to address these challenges have included the creation of synthetic datasets¹⁵, which simulate real-world scenarios to augment training data, and the adoption of weakly supervised learning methods¹⁶ to minimize reliance on fully annotated datasets. However, these approaches often fail to achieve the robustness and generalizability required for practical applications in blueberry production⁷. Issues such as occlusions, irregular fruit structures, and overlapping clusters exacerbate the difficulty of dense blueberry segmentation, underscoring the need for more comprehensive solutions.

Wang *et al.*¹⁷ proposed a depth-aware approach to detect concealed crops in dense fields, utilizing depth information and deep learning to overcome the challenges of overlapping vegetation. Similarly, Wang *et al.*¹⁸ improved the DLv3+ algorithm for high-resolution remote sensing images, introducing custom post-processing steps to improve segmentation accuracy. Despite these algorithmic refinements, challenges such as dense object overlap and occlusion remain inadequately addressed for blueberry applications. In addition to transformer-based approaches, traditional CNN-based methods continue to play a role in agricultural segmentation. Maheswari *et al.*¹⁹ developed an intelligent yield estimation framework using the SegNet (SN) model²⁰ for tomato crop segmentation, achieving high precision. However, its dependence on specific architectures restricts its flexibility for other crops, including blueberries. Kamal *et al.*²¹ employed a framework based on fully convolutional networks (FCN)²² for the segmentation of weeds and crops in agriculture aided by the Internet of Things (IoT). Although effective for binary segmentation, this approach performs poorly in multiclass scenarios, such as differentiating ripeness stages in dense blueberry clusters. Comparative studies have also highlighted the strengths and limitations of various CNN architectures. Abd Almisreb *et al.*²³ evaluated CNN-based models for plant leaf disease segmentation, providing insights into their effectiveness. However, the study did not explore advanced transformer-based architectures or hybrid models that integrate CNNs with other deep learning paradigms for blueberry applications.

Potential Research Gaps and Motivation. Recent innovations in computer vision techniques, including transformers and advanced CNN architectures, have significantly improved the detection of densely distributed objects in various domains^{5,6}. Despite this progress, the application of existing AI models to dense blueberry segmentation presents substantial challenges, primarily due to the unique complexities of agricultural environments and the lack of diverse large-scale annotated datasets that reflect real-world conditions. The limitations of existing datasets and the performance degradation of models due to data scarcity highlight the need for innovative solutions. This study aims to address these gaps by introducing a large-scale data repository explicitly designed for dense blueberry segmentation in production environments. The repository consists of a benchmark dataset with precise manual annotations, a semi-annotated dataset to facilitate weakly and semi-supervised learning, and a synthetic dataset generated using a novel diffusion algorithm to replicate real-world complexities. These datasets are intended to provide a robust foundation for advancing blueberry segmentation techniques. Furthermore, this study benchmarks the performance of state-of-the-art models using the proposed datasets and introduces an efficient detection framework optimized for densely clustered blueberries.

Contribution. This study addresses the challenges associated with dense blueberry segmentation in production environments and provides new foundations for the advancement of this research through the following key contributions:

- To advance AI-powered robotic vision for blueberry segmentation, three datasets were introduced: a benchmark for supervised learning, a semi-annotated dataset for weakly and semi-supervised learning, and a synthetic dataset replicating real-world complexities. These resources establish a strong foundation for blueberry segmentation advancements and domain adaptation.
- We developed a novel synthetic data generation algorithm that provides a scalable and cost-effective solution to generate annotated datasets, adaptable to blueberries and potentially other similar densely clustered fruits, advancing research on unsupervised learning, automated annotation, and domain adaptation.
- We benchmarked our dataset using a strong baseline and proposed an efficient framework optimized for densely clustered blueberries, achieving significant improvements in robustness and precision.
- To accelerate future research and innovation in precision agriculture and AI-driven segmentation, we have made the complete framework, including annotated datasets (DB-1, DB-2, DB-3), publicly available in the FigShare repository²⁴.

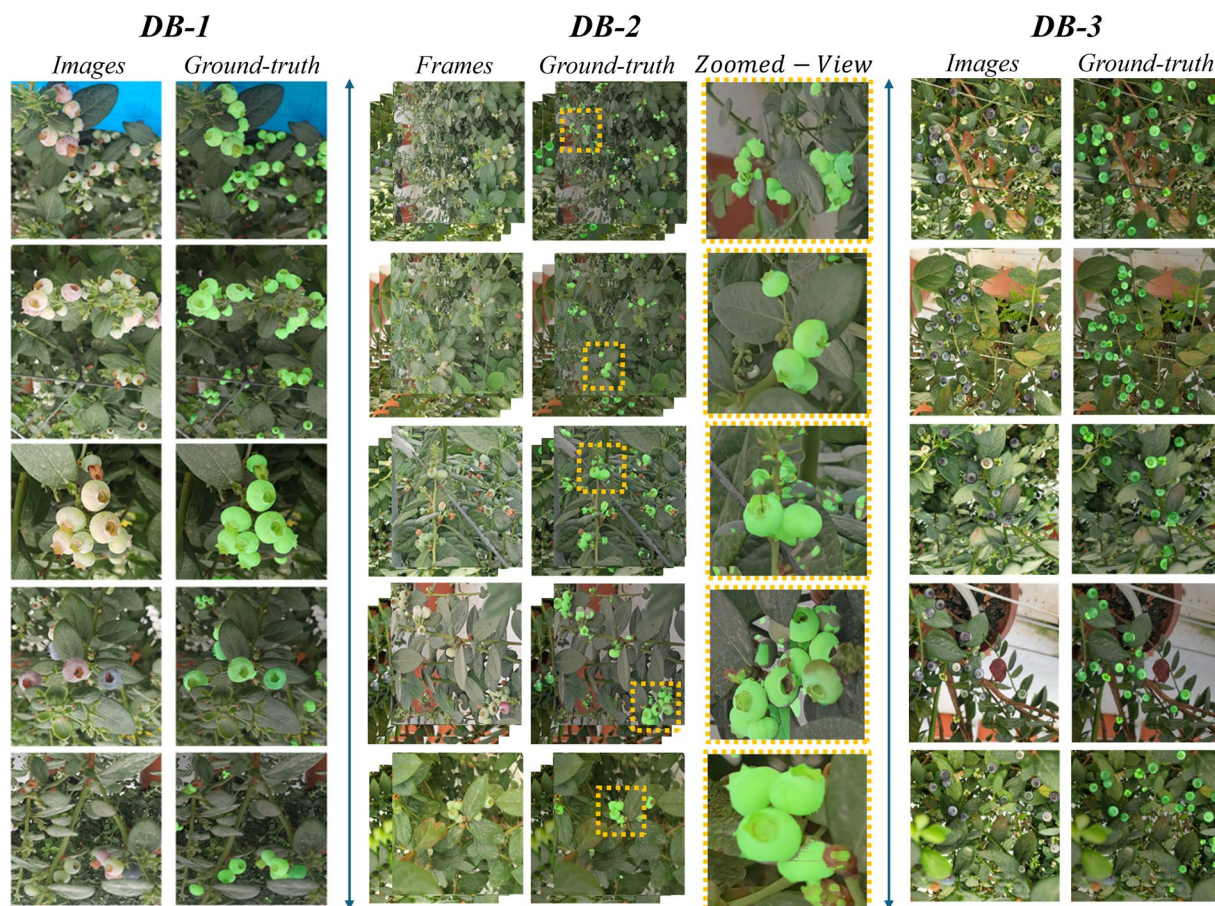


Fig. 1 Sample images from the proposed datasets with corresponding ground truth annotations: DB-1 (benchmark dataset for supervised learning), DB-2 (semi-annotated noisy dataset for weakly and semi-supervised learning), and DB-3 (synthetic dataset simulating real-world complexities for unsupervised learning and domain adaptation).

- The remainder of this paper is organized as follows: Methods section introduces the datasets and the proposed data generation model. The next section describes the data record. Finally, the Technical Validation section outlines the validation framework and the obtained results.

Methods

In this study, we present three distinct datasets (Fig. 1) to advance research and development in dense crop segmentation, with a focus on blueberries. Each dataset contributes unique characteristics and opens new avenues for addressing key challenges in dense crop segmentation to foster innovation in precision agriculture and related applications, as summarized in Table 3.

DB-1: Benchmark Dataset for Supervised Learning. The dataset (DB-1) was collected in a commercial greenhouse operated by Silal in Al-Foah, UAE, under production-scale conditions. The natural variability in illumination resulted from filtered sunlight, structural shadows, and heterogeneous artificial lighting. Four mobile cameras captured the dataset: 1) Samsung Galaxy S23 Ultra (200 MP, 12 mm f/1.7), 2) iPhone 14 Pro Max (48 MP, 24 mm f/1.8), 3) Huawei P50 Pro (50 MP, 23 mm f/1.8), and 4) Oppo Find X5 (50 MP, 24 mm f/1.8). A total of 520 videos (20–45 seconds each) were recorded with slow panning motions at 0.5–1.5 m canopy distances, yielding 141,453 frames at resolutions from 3024×4032 to 4000×6000 pixels. From these videos, a subset of 1,195 keyframes was selected for manual annotation. Three trained annotators manually labeled the frames, with cross-validation and consensus applied for challenging cases involving occlusion, overlapping clusters, and partial visibility. The annotated berries averaged 240–350 pixels in coverage, varying with camera distance and berry size. This process yielded precise annotations for 51,145 individual blueberries, capturing a wide variety of scenarios, including occlusions, overlapping clusters, and various environmental conditions. DB-1 is a high-quality benchmark dataset created specifically for dense crop segmentation tasks in real-world agricultural environments.

The complete dataset was collected on March 27, 2024, between 12:00 pm and 4:00 pm. The study focused on a single high-bush blueberry variety with mature bushes (1.0 to 1.5 m canopy height) grown in medium-sized pots (15 to 20 liters). The fruits were harvested in various stages of natural ripeness, with environmental

Aspect	Details
Location	Commercial indoor greenhouse, Silal, Al-Foah, UAE (production-scale setup).
Crop	Highbush blueberry, single variety, mature bushes grown in medium-sized containers (15-20 L) with a canopy height of 1.0-1.5 m; average berry diameter 15-20 mm; plants aged 2-2.5 years within an 8-10 year lifecycle.
Cameras	1) Samsung Galaxy S23 Ultra (200 MP, 12 mm f/1.7), 2) iPhone 14 Pro Max (48 MP, 24 mm f/1.8), 3) Huawei P50 Pro (50 MP, 23 mm f/1.8), 4) Oppo Find X5 (50 MP, 24 mm f/1.8).
Acquisition Time	March 27, 2024, between 12:00-16:00, capturing natural variability in lighting.
Lighting & Environment	Filtered sunlight through panels, structural shadows, heterogeneous artificial lighting, and leaf occlusion.
Videos	520 videos (20-45 s each) recorded with slow horizontal/vertical sweeps at canopy distances of 0.5-1.5 m.
Frames & Resolution	141,453 frames, resolution range 3024 × 4032 to 4000 × 6000 pixels.
Keyframes Annotated	1,195 frames, precisely annotated (polygon-based). 51,145 individual blueberries segmented.
Annotation Details	Three trained annotators; two-stage validation with consensus for occluded/overlapping fruits. Average berry size: 240-350 pixels.
Limitations	Single cultivar, greenhouse-only setting, subjectivity in ripeness labeling.

Table 1. Summary of DB-1 dataset acquisition, including environment, hardware, conditions, and annotation details.

variability, including lighting differences, natural shading, and leaf occlusion preserved to enhance ecological validity. Camera movements incorporated slow horizontal and vertical sweeps to document structural complexities such as dense clusters, overlapping fruits, and foliage interference. A structured summary of dataset acquisition is provided in Table 1.

To ensure its effectiveness as a reference dataset, DB-1 was evaluated in a fully supervised learning environment. We divide the data into 70% for training, 10% for validation, and 20% for testing. We benchmarked our dataset using a strong baseline and proposed a customized framework specifically to address the unique challenges of dense crop segmentation in wild environments. Our framework demonstrated superior performance compared to existing state-of-the-art methods.

The annotation of DB-1 was carried out through a structured multi-step protocol to ensure both accuracy and consistency, particularly for densely clustered and small-sized fruit. Each annotated frame was subjected to a two-stage validation process involving three trained annotators. In the first stage, annotators independently labeled frames with careful attention to boundary delineation. The initial segmentation masks were drawn manually using a polygon-based labeling tool, focusing on a clear boundary separation for each individual blueberry. Special attention was given to cluttered regions, where fruits often overlapped or exhibited partial occlusion. In the second stage, annotations were reviewed for consistency and completeness, with a particular focus on challenging regions such as overlapping clusters, heavily shadowed fruits, or partial visibility. Discrepancies were resolved through consensus discussion among the annotators until full agreement was reached, minimizing subjective bias. This consensus-based protocol was particularly important to ensure high-quality masks in challenging scenes, which are critical to training and evaluating supervised segmentation models. The complete structured annotation workflow is also summarized in Table 2. Although formal quantitative inter-annotator agreement metrics were not collected during the annotation process, the structured two-stage validation with consensus review ensured consistent boundary delineation, and annotation quality is further evidenced by the reliable performance of multiple state-of-the-art models trained on this dataset. The resulting masks offer high spatial accuracy and instance-level separation, making DB-1 a robust benchmark for training and evaluating segmentation models in real-world agricultural settings.

DB-2: Semi-Automated Noisy Annotation for Video Data. DB-2 expands on the initial dataset by annotating the remaining 141,453 frames extracted from the collected videos. Unlike DB-1, which provides high-precision manual annotations, DB-2 employs a semi-automated annotation process. Specifically, our proposed segmentation framework trained on DB-1 was used to infer segmentation masks for the remaining video frames, which naturally introduces label noise but enables cost-effective scaling to the full dataset. The videos were captured using the same acquisition protocol as DB-1, with slow panning motions (horizontal and vertical sweeps) at canopy distances of 0.5 to 1.5 m, ensuring coverage of dense clusters, occluded fruits and varying perspectives across greenhouse rows. Although this semi-automated approach results in less accurate labels compared to manual annotation, it mirrors real-world scenarios where large-scale annotation is often limited by time and resource constraints. Despite these limitations, DB-2 serves as a valuable resource for developing robust AI models capable of handling annotation imperfections and variability. It supports future applications such as few-shot learning, where models are trained to generalize using minimal labeled data, and weakly supervised learning, which leverages noisy annotations to reduce dependence on clean, large-scale labels. Additionally, DB-2 enables semi-supervised learning approaches that combine clean labels from DB-1 with noisy labels from DB-2 to enhance segmentation performance.

DB-3: Synthetic Dataset. To mitigate the high costs and labor intensive nature of manual annotation, we developed a synthetic dataset (DB-3) using a custom synthetic data generation algorithm (outlined in Table 4).

Stage	Description
Stage 1: Initial Annotation	Each annotator $a_j \in \{a_1, a_2, a_3\}$ independently labels frame I_n , producing a polygon-based mask $M_i^{(j)}$ with boundaries ∂B_k for each blueberry B_k . Special focus is given to overlapping and partially occluded fruits.
Stage 2: Review and Validation	Masks are reviewed for completeness and consistency. Ambiguous regions Ω (occlusions, shadows, partial visibility) are identified by comparing $M_i^{(j)} \Delta M_i^{(k)}, \forall j \neq k$.
Stage 3: Consensus Resolution	Annotators resolve discrepancies via consensus discussion, producing the final mask: $M_i^* = f_{\text{consensus}}(M_i^{(1)}, M_i^{(2)}, M_i^{(3)})$ ensuring minimized subjectivity.
Stage 4: Final Dataset	Final masks $\{M_i^*\}$ are stored with one-to-one correspondence to images. Total: 51,145 annotated blueberries, average size ≈ 240 -350 pixels.

Table 2. Structured two-stage annotation protocol for DB-1.

Dataset	Samples	Annotation Type	Significance	Applications	Pros and Cons
DB-1	1,195 images	Manual expert annotations	Provides highly accurate labels for dense crop segmentation	Used as a benchmark for supervised learning and model evaluation	Pros: Reliable annotations, ideal for training. Cons: Time-intensive, small dataset size.
DB-2	141,453 frames from 520 videos	Semi-automated annotations	Large-scale dataset with realistic variability for generalization tasks	Facilitates weakly supervised, semi-supervised, and noisy label adaptation methods	Pros: Scalable, reflects field variability. Cons: Labels may introduce noise, impacting precision.
DB-3	10,000 synthetic images	Algorithm-generated annotations	Cost-effective alternative offering diverse conditions, including occlusions and clutter	Supports unsupervised learning, domain transfer, and few-shot learning	Pros: Scalable, reduces manual annotation costs. Cons: Synthetic data may not fully replicate real-world complexities.

Table 3. Summary of the three datasets (DB-1, DB-2, and DB-3) introduced in this study to address key challenges in dense crop segmentation and foster innovation in precision agriculture, particularly focused on blueberries.

This algorithm incorporates image blending and fusion techniques to simulate real-world complexities, such as overlapping clusters, occlusions, and varying fruit sizes. It combines multiscale transformations, randomized placements, and adaptive blending methods to generate realistic images and corresponding segmentation masks by integrating leafy background images with blueberry patches. The resulting dataset comprises 10,000 synthetic images, each paired with highly accurate segmentation masks. By introducing additional complexities, such as cluttered objects and overlapping berries, DB-3 offers a controlled but challenging environment to evaluate model performance and extends the scope of research on dense crop segmentation by enabling various learning paradigms.

To better simulate real-world challenges, certain visual artifacts (e.g., partially cut fruits, boundary truncation, halos, and shading variations) were deliberately introduced in DB-3. These artifacts mimic natural conditions, such as occlusions, overlapping clusters, incomplete fruit visibility, and irregular illumination, which are common in dense crop environments but often absent in artificially clean datasets. Our pipeline is intentionally training-free, relying only on a small set of fruit patches and leaf images to generate both synthetic images and perfectly aligned pixel-level segmentation masks simultaneously. In contrast, GAN-based or diffusion-based generative methods require large annotated datasets for training and typically produce images without the corresponding segmentation masks, which limits their direct applicability to segmentation tasks. Our approach prioritizes scalability, reproducibility, and automatic mask generation, making it accessible to researchers with limited resources. Although not employed here, future extensions may explore hybrid approaches that combine our lightweight training-free pipeline with generative refinement techniques to further enhance photorealism while maintaining automatic mask generation capability.

Although the synthetic data cannot fully replace original field images, it offers two main benefits: 1) providing a scalable and cost-effective way to pre-train models before fine-tuning with a small number of real annotated samples, and 2) establishing a reusable pipeline that can be adapted to other crops, reducing the burden of exhaustive manual labeling. In unsupervised learning, synthetic data serve as a training resource without requiring explicit labels, facilitating advancements in domain adaptation. The dataset also supports domain adaptation techniques, enabling the transfer of knowledge from synthetic to real-world scenarios, thereby bridging the gap between simulated and real agricultural environments. Furthermore, DB-3 can be leveraged for few-shot refinement, where a small portion of real-world data is used to fine-tune models pre-trained on synthetic data. The dataset promotes robust model training by introducing variations and occlusions, improving the resilience of the model to challenges encountered in real-world agricultural settings.

Data Record

Table 5 outlines the complete folder structure, file-naming patterns, and dataset contents. All images are stored in lossless formats to retain original quality and ensure compatibility with common supervised learning workflows. The dataset is organized into three main directories: DB-1, DB-2, and DB-3. The complete dataset, including original frames, annotated masks, and synthetic images, is hosted on the *Figshare repository* and can be accessed in the *FigShare repository*²⁴.

Step	Description and Equations
0	Initialize masks:
	$M_u(x, y) \leftarrow 0, M_c(x, y) \leftarrow 0$ (zero matrices matching background dimensions).
1	Initialize parameters:
	Inputs:
	M number of fruit images $\mathcal{F} = \{F_1, \dots, F_M\}$.
	N number of leafy background images $\mathcal{B} = \{B_1, \dots, B_N\}$.
	Parameters:
	$bmin, bmax$: Minimum/maximum berries per image (typically 15–50).
	$smin, s_{max}$: Scaling range for size diversity (typically 0.5–1.5).
	cp : Percentage of berries with partial occlusion (typically 0.3–0.4).
	σ_{blur} : Gaussian blur radius for edge smoothing (typically 3–5 pixels).
	α : Blending weight for berry opacity (typically 0.7–0.9).
2	Select background:
	Randomly choose $B \in \mathcal{B}$. Identify green regions $G(x, y)$ using HSV thresholds ³⁹ .
3	Choose blueberry count:
	$b \sim \text{uniform}(bmin, bmax), c = \lfloor c_p \cdot b \rfloor$ (number with occlusion).
4	Process each blueberry F_j for $j = 1, \dots, b$:
	- Sample scale and rotation: $s \sim \text{uniform}(smin, s_{max}), \theta \sim \text{uniform}(0, 360)$.
	- Transform berry: $F' = \text{rotate}(\text{scale}(F_j, s), \theta)$.
	- Sample center: $(x_c, y_c) \sim G$ (random location within green regions), $r = s \cdot r_0$.
	- Generate circular mask: $C_j(x, y) = 1$ if $(x - x_c)^2 + (y - y_c)^2 \leq r^2$, else 0.
	- For cluttered berries ($j \leq c$): $C'_j = C_j \cdot \text{partialMask}$ (random occlusion pattern).
5	Accumulate segmentation masks:
	Uncluttered ($j > c$): $M_u(x, y) \leftarrow \max(M_u(x, y), C_j(x, y) \cdot 255)$.
	Cluttered ($j \leq c$): $M_c(x, y) \leftarrow \max(M_c(x, y), C'_j(x, y) \cdot 255)$.
6	Blend into background:
	Uncluttered: $B_u(x, y) \leftarrow \alpha F'(x, y) + (1 - \alpha)B(x, y)$ where $C_j = 1$.
	Cluttered: $B_c(x, y) \leftarrow \alpha F'(x, y) + (1 - \alpha)B(x, y)$ where $C'_j = 1$.
7	Apply edge smoothing:
	Smooth blended images with Gaussian blur:
	$B'(x, y) = \text{GaussianBlur}(\text{Dilate}(\text{Canny}(B_u, M_u), rd), \sigma_{blur})$.
8	Save results:
	Save B', M_u to uncluttered folders; B', M_c to cluttered folders.

Table 4. Detailed overview of the synthetic dataset generation algorithm developed in this study.

DB-1. Contains a manually selected subset of frames, extracted at fixed intervals (every 60 frames) from videos recorded by four mobile cameras. This directory includes two subfolders: *Images*, which holds the extracted image frames, and *Masks*, which holds their corresponding segmentation masks. All files use the naming convention *Camera_x_Video_y_z*, where x is the camera ID, y is the video number, and z is the frame index. The image files are stored in .jpg format, while masks use the .png extension. In total, this directory contains 1,195 image-mask pairs.

DB-2. Provides a complete set of frames from the same video recordings, without downsampling. The data is split into two folders: *Video_Frames* for the raw image frames and *Video_Frames_Masks* for the corresponding pixel-wise segmentation masks. The filenames follow the same structured format as used in DB-1. This directory contains a total of 141,453 frames, with an equal number of annotated masks, covering all four camera views.

DB-3. Consists of synthetically generated samples created for the purpose of data augmentation and evaluation of model generalization. This part of the dataset includes *Generated_Images* and *Generated_Masks*. Filenames follow a simple sequence-based convention: *synthetic_image_x*, where x is a unique index. Image files use the .jpg extension, and masks use the corresponding .png format to ensure one-to-one pairing.

Data Limitations. Although DB-1 is designed as a high-quality benchmark, certain limitations are acknowledged: 1) The data were collected on a single day (27 March 2024) under controlled greenhouse conditions, ensuring consistency, but limiting exposure to natural variations such as outdoor lighting and seasonal effects. 2) The dataset focuses on a single blueberry cultivar at a maturity stage, which restricts varietal and architectural diversity, while manual annotation, despite consensus validation, introduces minor subjectivity in ripeness categorization. 3) Furthermore, data were captured using high-resolution smartphone cameras, which, although practical and accessible, do not include other modalities such as multispectral or robotic systems. However, these

Dataset	Folder Path	Description and Structure
DB-1	DB-1/Images	Contains selected frames sampled every 60 frames. Naming format: Camera_x_Video_y_z.jpg. Where x = camera number, y = video number, z = frame index. Total: 1 195 images across 4 cameras. <i>Example: Camera_1_Video_5_61.jpg.</i>
	DB-1/Masks	Contains segmentation masks aligned with each image. Naming mirrors image filenames with extension .png. <i>Example: Camera_1_Video_5_61.png.</i> Total: 1 195 masks (one-to-one correspondence).
DB-2	DB-2/Camera#/Video_Frames	Contains all frames from videos at full sampling rate. Naming format: Camera_x_Video_y_z.jpg. Total frames: 141 453 (Camera1 31,419; Camera2 14,792; Camera3 67,377; Camera4 27,865). <i>Example: Camera_3_Video_126_5609.jpg.</i>
	DB-2/Camera#/Video_Frames_Masks	Masks for every frame in Video_Frames. Naming identical to frame filenames with .png extension. Total: 141 453 masks (perfect one-to-one mapping). <i>Example: Camera_3_Video_126_5609.png.</i>
DB-3	DB-3/Generated_Images	Contains synthetically generated images. Naming format: synthetic_image_x.jpg, where x is a sequential index. <i>Example: synthetic_image_42.jpg.</i>
	DB-3/Generated_Masks	Masks corresponding to each synthetic image. Naming matches image filenames with .png. <i>Example: synthetic_image_42.png.</i> One-to-one correspondence maintained.

Table 5. Overview of the dataset directory structure, including folder organization, file naming conventions, and image-mask pairing details for DB-1, DB-2, and DB-3.

constraints do not significantly affect the reliability of the dataset as a benchmark: controlled setup ensures reproducibility, smartphone acquisition improves accessibility, and the consensus-driven protocol delivers high-quality segmentation masks even in cluttered regions. DB-1 therefore provides a solid and reproducible foundation for validating AI-based dense fruit segmentation, with future extensions (e.g., multiple cultivars, outdoor conditions, and alternative sensors) expected to broaden its generalizability.

Future Directions. This work opens multiple avenues for the advancement of dense crop analysis using the three proposed datasets. Accurate segmentation enables extensions to berry counting and yield estimation. Future efforts will target robust counting under varying lighting and growth conditions through temporal tracking and multi-view fusion. For the synthetic dataset (DB-3), systematic validation will be prioritized in future work by training on synthetic data and evaluating on real or hybrid sets to assess the effectiveness of the synthetic data provided as an enhancement resource. Additional directions include integrating transfer learning to reduce domain gaps, developing weakly and self-supervised methods using semi-annotated and synthetic datasets, and extending the current algorithm (designed for dense blueberry segmentation) to other crops such as grapes, strawberries, tomatoes, and citrus²⁵. Finally, adapting the algorithm for drone-based synthetic annotation¹⁶ with multi-sensor data fusion can better replicate real-world complexities and support high-resolution, real-time agricultural analysis²⁶.

Technical Validation

To assess its technical quality and applicability, we benchmarked our dataset using a strong baseline and proposed an efficient framework optimized for densely clustered blueberries. These experiments validate the consistency and usability of the annotated data in real-world scenarios with densely clustered fruit, establishing DB-1 as a robust supervised learning benchmark. Our present focus is to provide the community with three complementary datasets, with DB-1 serving as the validated foundation and DB-2/DB-3 offering future directions. The following sections describe the model, the training strategy, and the results obtained in DB-1, establishing reproducible baselines for future research.

Model Development. We developed a domain-specific segmentation framework. The framework integrates two complementary networks: 1) the Pyramid Vision Transformer Segmentation Network (PVT-SN) and 2) the Segment Anything Model Refinement Network (SAM-RN). The PVT-SN leverages transformer-based hierarchical feature extraction and pyramidal pooling to capture both global context and local detail^{27,28}, producing an initial segmentation mask. The transformer backbone captures long-range dependencies, enabling better boundary disambiguation in cluttered regions, while the reshaping operation fuses local and global features for improved precision. The SAM-RN, built on the Segment Anything Model²⁹, refines this mask using bounding box prompts derived from the PVT-SN output to improve boundary precision and recover missed detections in cluttered or occluded regions. Finally, a post-fusion stage combines the predictions of PVT-SN and SAM-RN through a union operation, ensuring high sensitivity without compromising specificity. The combined design takes advantage of the modeling of the lobe characteristics of the PVT subnetwork, the robust zero-shot refinement of the SAM, and the post-fusion strategy to address the unique challenges of densely fruited crops, including clusters overlapping and heavy occlusions. Although these models represent strong architectural choices, they are used primarily here to establish reproducible benchmarks on DB-1; future research can extend or replace them with alternative architectures. For detailed workings of the ViT and SAM components, readers are referred to their original publications^{27,29}. The overall design and workflow are illustrated in Fig. 2.

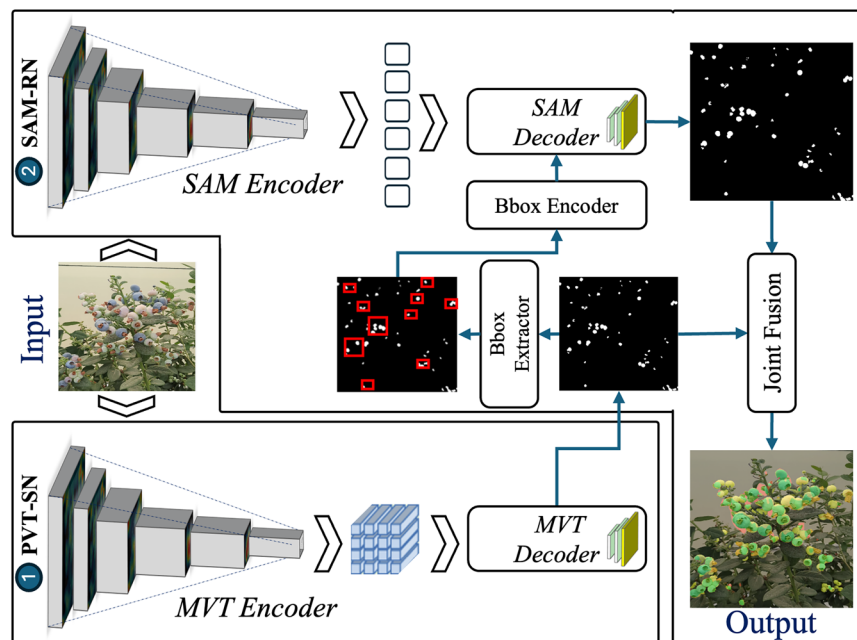


Fig. 2 Overall model design and workflow diagram of the proposed framework for dense crop segmentation, comprising two core networks: 1) the Pyramid Vision Transformer-Segmentation Network and 2) the Segment Anything Model Refinement Network.

Model	SEN (%)	SPE (%)	PRE (%)	IoU (%)	DICE (%)
VT-SN ⁵	60.19	98.15	74.91	50.09	66.75
PVT-SN	<u>69.74</u>	<u>97.55</u>	<u>72.30</u>	<u>55.03</u>	<u>71.00</u>
PVT-SN+SAM-RN	75.06 ↑7.63%	97.06↓1.11%	70.09↓6.44%	56.85 ↑3.31%	72.49 ↑2.10%

Table 6. Quantitative ablation study showing incremental performance gains with the addition of modules to the baseline model, resulting in the final customized model for dense crop analysis. **Bold: Best results**, arrows indicate relative improvement (↑) or decrease (↓) of proposed model compared to second-best or best value respectively.

Validation Setup. To evaluate the performance of the proposed model specified in a fully supervised environment, the DB-1 dataset was divided into 70% for training, 10% for validation, and 20% for testing. The current validation focuses on establishing DB-1 as a supervised learning benchmark. Although this demonstrates the effectiveness of the dataset for supervised segmentation tasks, future validation studies that include domain transfer experiments from synthetic data (DB-3) to real data, systematic evaluation of semi-supervised learning performance using DB-2, and cross-dataset performance analysis would further demonstrate the utility of our complete dataset repository. The experiments were carried out using MATLAB R2024a with its deep learning toolbox on a high performance system featuring an Intel Core i9 processor, 128 GB RAM, and an NVIDIA RTX A6000 GPU (48 GB VRAM), having Windows 11. Key segmentation metrics, including Sensitivity (SEN), Specificity (SPE), Precision (PRE), Intersection over Union (IoU), and DICE Coefficient (DICE), were used to comprehensively assess model performance, equipping a stringent comparison to baseline models. noindent **Hyperparameters:** The main hyperparameters were the following: learning rate 1×10^{-4} , Adam optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.999$, batch size of 12, patch size of 16×16 , embedding dimension $d = 768$, number of transformer layers $N = 12$, attention heads $h = 12$ and dropout rate of 0.1. For SAM-RN, we used its lightweight ViT backbone with patch size 16×16 , embedding dimension $d = 256$, and 12 attention heads. Early stopping (patience = 15) and a maximum of 100 training epochs were applied.

Model Training. The training process for the proposed framework employs a fully supervised approach for the PVT-SN model, while the SAM-RN model operates as a zero-shot model during the testing phase without requiring additional training. The PVT-SN network is trained using a weighted cross-entropy loss function³⁰, specifically designed to address the class imbalance commonly observed in dense crop segmentation tasks. The weighted cross-entropy loss is defined as:

$$\mathcal{L}_{WCE} = - \sum_j (\alpha M_i[j] \log(\hat{M}_{PVT-SN,i}[j]) + \beta (1 - M_i[j]) \log(1 - \hat{M}_{PVT-SN,i}[j])) \quad (1)$$

where $M \in \{0, 1\}^{H \times W}$ is the mask of the ground truth, $\hat{M}_{PVT-SN} \in [0, 1]^{H \times W}$ is the predicted mask, and α, β are adaptive weights that balance the contributions of the foreground and background. These weights are computed

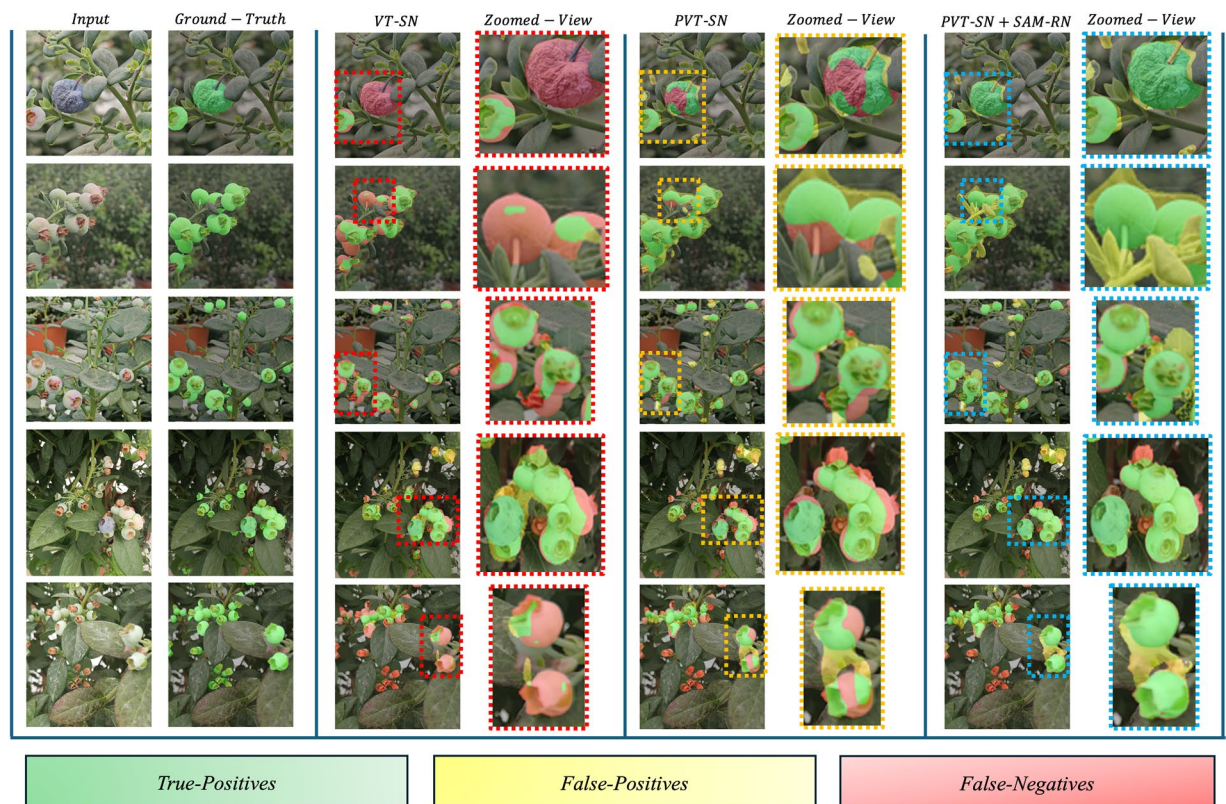


Fig. 3 Visual comparison of segmentation results for the baseline VT-SN, PVT-SN, and the final PVT-SN+SAM-RN framework. Zoomed patches highlight the incremental improvements in segmentation accuracy at each stage.

dynamically for each training batch based on the inverse frequency of the foreground and background pixels, normalized to sum to 1. This adaptive weighting strategy automatically handles class imbalance without requiring manual hyperparameter tuning, ensuring the model emphasizes under-represented fruit regions across varying image conditions. The integration of the transformer backbone (PVT) enables global context modeling and long-range dependency capture, which proves essential for disambiguating overlapping boundaries in densely clustered scenarios where local receptive fields alone cannot distinguish between partially occluded berries and background foliage. The reshaping operation bridges local convolutional features with global transformer features, refining predictions in heavily cluttered regions.

Results. To evaluate the proposed model, a three-stage ablation study was conducted. The baseline Vision Transformer-Based Segmentation Network (VT-SN), implemented using SegFormer⁵, was chosen for its proven efficiency in general³¹, medical^{32–34}, and agricultural segmentation tasks^{12,35,36}. The framework was then enhanced to PVT-SN by adding a pyramid-pooling block and further extended with SAM-RN (PVT-SN+SAM-RN) to better address occlusion and clutter in dense scenes. The baseline VT-SN achieved 66.75% DICE and 50.09% IoU. Adding a Pyramid-Pooling block produced the PVT-SN model, which captured multiscale spatial features and improved segmentation accuracy to 71.00% DICE and 55.03% IoU—relative gains of 6.37% and 9.86%, respectively. SEN also increased from 60.19% to 69.74% (+15.87%), showing stronger detection of dense fruit clusters vital for estimating yield and ripeness^{1,37}. Incorporating SAM-RN into the second stage further reduced false negatives, increasing SEN to 75.06% (+7.63%), with additional improvements to 72.49% DICE and 56.85% IoU, relative gains of 2.10% and 3.31% over PVT-SN.

The final PVT-SN+SAM-RN framework significantly outperforms the baseline VT-SN, achieving relative gains of 24.71%, 13.5%, and 8.6% in SEN, IoU and DICE, respectively. Minor trade-offs occur in specificity (98.15 → 97.06%) and precision (74.91 → 70.09%) due to a stronger focus on minimizing false negatives, slightly increasing false positives. Despite this, the model maintains an effective balance between sensitivity and specificity, ensuring accurate segmentation under dense and occluded conditions. These quantitative results are summarized in Table 6, highlighting the incremental improvements achieved through each design stage. Statistical analysis using a paired t-test³⁸ confirmed the improvements as significant ($p < 0.01$, 99% confidence). Figure 3 further illustrates this progression: VT-SN shows incomplete fruit boundaries and missed detections, PVT-SN refines coverage with clearer boundaries, and the final PVT-SN+SAM-RN recovers occluded fruits and enhances sensitivity. The visual and quantitative results together validate the robustness and reliability of the proposed model for precise dense crop segmentation.

Data availability

The datasets supporting this work, including DB-1, DB-2, and DB-3 along with their corresponding annotations, have been deposited in the Figshare repository. All data are publicly available and can be accessed at²⁴.

Code availability

The complete algorithm for data generation is provided in this paper. The datasets (DB-1, DB-2, and DB-3), along with annotation files and supporting documentation, are also released to support reproducible research and further development in dense crop analysis and AI-driven agricultural segmentation. All resources are hosted on the Figshare repository and can be accessed at²⁴.

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Author contributions

Conceptualization, Data curation, and Methodology, M.O.; Investigation and Validation, M.S., M.Z.; Writing - Original draft, M.O.; Data curation and Resources, S.M., Writing—review and editing, M.O. and I.H.; Supervision, Funding acquisition, and Resources, L.S., and I.H.; All authors have read and agreed to the currently submitted version of the manuscript.

Competing interests

The authors declare no competing interests.

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